Model Specialization for Causal Programs

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April 14, 2026

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1 Motivation

General statistical inference is a powerful tool for modelling the randomness of our universe, but does not always translate clearly into the modelling of programmatic procedures that dynamically observe, interact with, make decisions based on, and even modify that randomness. Such "causal programs" are not well understood by the state of the art.

This report outlines a framework for extending statistical modelling to reason about causal programs with efficient computation and formal specification.

You may skip to sections 2 for a summary of the $\mathbb{AIM}$ semantics used to formalize causal programs, 3 for the algebraic overview the bridges the gap between syntax of causal programs and creation of "procedures-to-claim" proofs that certain procedures suffice to prove certain claims, 4 for a concretization of how this algebra translates into proof obligations, or to 5 to ground all this high-level wizardry in practical examples.

1.1 Failures of General Inference

In the modern age, ridiculously high-dimensional "overparameterized" inference is broadly considered tractable with two important caveats: *explainability* and *representability*.

First, inference is only answered in an approximate black box sense; given an arbitrarily large set of existing samples, arbitrarily large sets of fresh samples can be generated, but insight into how or why that particular "fit" was obtained are not forthcoming. This is often referred to as a failure of *explainability*.

Second, not all problem domains are easily mapped to well-understood statistical models. This can produce a failure of *representability*. In some domains, such as video comprehension, clever "featurization" provides this mapping. In others, such as *causal* inference, the mapping relies on subtle, untestable assumptions. The results of inference can be very unstable with respect to these assumptions: small logical errors in the representation of problems can render the resultant analyses completely meaningless.

Unfortunately, in many domains, the parallel goals of explainability and representability are fundamentally opposed. For example, increasing the dimension of parameterization comes at a nearly universal increase in representability at the cost of explainability. Other choices, such as forcing candidate overparameterized models to correspond to simpler "underparamaterized" models with human-readable semantics ensures some explainability at the cost of some representability.

1.2 Partial Solution: Structural Causal Models

Structural Causal Models provide a powerful intersection of explainability and representability in the domain of causal inference. In particular, a *causal structure* is assumed that constrains the dependence of observable quantities of interest to a small set of "parents"; consisting of a mix of other observable values and hidden/latent "unobservable" values. The dependence of each such causal node on its parents is explainable as a small, simply-parameterized model, and the remaining signal that cannot be thus modelled is shunted into raw statistical inference on the unobserved values.

Causal models thus enforce a powerful "separation of concerns": the explainable causal dependencies of observable quantities on each other are parameterized seperately from the remaining unexplainable statistical dependencies of those quantities on universal randomness. The parameterization of the model is thus split into the former *causal parameters* and the latter *statistical parameters*.

Thus purely statistical operations (like sampling fresh units, conditioning on observable properties of units, and observing properties of units), and causal operations ("interventions" that alter the causal mechanisms at play) are equally expressible in structural causal models.

To this extent, structural causal models can be said to faciliate a nearly-complete reduction of causal inference to general statistical inference.

1.3 Complete Solution: Specialization to Causal Programs

Unfortunately, directly reducing the problem of causal inference to the synthesis and use of statistical "black boxes" is often unsatisfactory. All the problems of general inference lift through this reduction to become properties of general causal inference: ridiculously high dimensionality, limited explainability, and instability with respect to subtle assumptions. Most crucially, it is nearly impossible to establish that the inferred black boxes adhere to any meaningful *specification* expressed in terms of the causal assumptions.

Our framework relies on simple formal descriptions of procedures for observing and interacting with the world ($\mathbb{AIM}$ programs), and models their behavior directly through an information-processing pipeline. Notable, each step of the pipeline has clearly defined semantics and specifications, yielding composed specifications applying to the results of inference through the pipleine.

Once built, this will allow specifications of strucutral causal models to be combined with specifications of data-gathering programs and query programs to sythensize or validate estimators that directly answer the query in terms of gathered data.

The estimator can be proven sound with respect to all assumptions and specifications with far greater precision that can be obtained directly through the application of general statistical inference.

2 Foundation: $\mathbb{AIM}$ Semantics for Causal programs

We define 4 modellings domains:

$\mathcal{U}$: domain of "unobservable states" - i.e. assignments to the unobservable statistical terms in the structural causal model

Θ : "causal state" - i.e. assignments of the particular relationships between observable terms and their causal parents

$\mathcal{R}$: "causal program" - the space of programs that may be modelled

$\mathcal{D}$: - the domain of "results"/"return values" of causal programs

Many design decisions influence the representation of these 4 domains. This report is motivated by a particular presentation of the domain $\mathcal{R}$ of causal programs consisting of $\mathbb{AIM}$ programs: programs consisting of three atomic operations:

$\mathbb{A}$: assign experimental conditions, equivalent to statistical conditioning

$\mathbb{I}$: intervene, equivalent to replacement or removal of causal mechanisms

$\mathbb{M}$: measure, which produces the observable outputs of the program)

Deferring to prior explanations of the exact semantics of $\mathbb{AIM}$ programs, it suffices to note for the sequel that a sufficient description of their semantics takes the form of inhabiting/deciding the relation:

$\boxed{\Theta, U \vdash \mathcal{R} \downarrow \mathcal{U}}$: "program $\mathcal{R}$ produces result $\mathcal{D}$ in the context of Θ and $\mathcal{U}$"

3 Framework: Information Processing Pipeline

Now assume that we have a semantics for causal programs $\mathcal{R}$ that deterministically produce outputs $\mathcal{D}$ in environments consisting of fixed $\mathcal{U}$ and Θ:

$$\boxed{\Theta, \mathcal{U} \vdash \mathcal{R} \downarrow \mathcal{U}}$$

Consider a partially Bayesian approach in which a natural prior p for $\mathcal{U}$ exists[1]. We can now consider $\mathcal{R}^{\{p\}}$ to be a probabilistic program that produces a *distribution* over outputs $\mathcal{D}$ for any fixed Θ:

$$\boxed{\Theta \vdash \mathcal{R}^{\{p\}} \downarrow \mathbb{P}_p[\mathcal{D}]}$$

This can be directly read as "program $\mathcal{R}^{\{p\}}$ produces results distributed according to $\mathbb{P}_p[\mathcal{D}]$ in the context of Θ when $\mathcal{U}$ is distributed according to p".

The details of efficiently translating representations of p and $\mathcal{R}$ into representations of $\mathbb{P}_p[\mathcal{D}]$ are subtle in the general case, but for small dimension, it is a simple "counting" problem: the probability associated with result $\mathcal{D}$ is the sum of all probabilities associated with $\mathcal{U}$ that map under Θ and $\mathcal{R}$ to $\mathcal{D}$.[2]

3.1 Experimental and Query Programs

Our key insight for specializing the model domain is to consider a *pair* of causal programs:

$\mathcal{R}_{exp}$: "experimental program" - describes the causal program used to gather data about the world; must correspond to a practical procedure that can be performed to build a dataset of outcomes given access to a large pool of individuals or other units to evaluate

$\mathcal{R}_{qry}$: "query program" - describes the causal program used to define the question about about the world we seek to answer; need not be practically executable itself

[1] see: max entropy priors

[2] in the general case, continuous and/or convex analysis or other approximation like MCMC may suffice

In general $\mathcal{R}_{qry}$ should be small and simple, so that predictions of its outcome are "human readable", but this may come at the cost of being practically realizable, but $\mathcal{R}_{exp}$ may be arbitrarily complex as needed to ensure correspondence to a realistic experimental procedure.

$$\begin{tikzcd} & \Theta \arrow[dl, "\mathcal{R}_{exp}^{\{p\}}"'] \arrow[dr, "\mathcal{R}_{qry}^{\{p\}}"] & \\ \mathbb{P}_p[\mathcal{D}_{exp}] & & \mathbb{P}_p[\mathcal{D}_{qry}] \end{tikzcd}$$

Figure 1: System of two probabilistic causal programs

Figure 1 illustrates a system by which choice of Θ induces distributions on the results each of these respective causal programs. Notably, these distributions may exhibit any degree of covariance. In general, increasing covariance represents increasing ability to infer one program from the other, and decreasing covariance represent increasing ability to efficiently represent the results of one program in terms of the other.

Many interesting properties are representable even in this simplistic domain. For example, if the resultant $\mathbb{P}_p$ distributions are related via a determinstic transform, we say that the query is "identifiable" from the experimental data.

3.2 "Soft" Posterior Estimators

We now make a slight alteration to the modelling domain to introduce the idea of "soft" estimates of the query program. In the modelling domain represented by figure 1, distributions on $\mathcal{D}_{qry}$ are not available unless a value of Θ is fixed. Alternatively, in the "fully Bayesian" case, if a prior on Θ itself can be fixed, then (joint) distributions of $\mathcal{D}_{exp}$ and $\mathcal{D}_{qry}$ are also fixed. Unfortunately, priors on Θ are not available or semantically meaningful in the general case.

However, if a result $\mathcal{D}_{exp}$ is fixed, then a *posterior* distribution on Θ, in which the liklihoods of various Θ are proportional to the relatively liklihood of our fixed $\mathcal{D}_{exp}$ under them, is also fixed. Thus, any observation of the result of $\mathcal{R}_{exp}$ fixes a posterior distribution on $\mathcal{D}_{qry}$ even without a prior on Θ:

Figure 2 depicts this relationship: fixing $\mathcal{R}_{exp}$ and $\mathcal{R}_{qry}$ induces a distribution on $\mathcal{D}_{qry}$ for any fixed observation of $\mathcal{D}_{exp}$. This induced mapping α is called a "soft" decision because it does not force choice of a particular value of $\mathcal{D}_{qry}$, but simply weights the relative liklihoods of each potential value of $\mathcal{D}_{qry}$ given the observation of $\mathcal{D}_{exp}$.

The diagram, and in particular the mapping under $\mathcal{R}_{exp}$ is mutilated accordingly to show that $\mathcal{R}_{exp}$, in this setting, is evaluated in its "backwards" direction rather than "forwards".

$$
\begin{tikzcd}
 & \Theta \arrow[dl, squiggly, "\mathcal{R}_{exp}^{\{p\}}"'] \arrow[dr, "\mathcal{R}_{qry}^{\{p\}}"] & \\
\mathcal{D}_{exp} \arrow[rr, dashed, "\alpha^{\{p\}}" description] & & \mathbb{P}_p[\mathcal{D}_{qry}]
\end{tikzcd}
$$

Figure 2: Posterior-liklihood characterization of two programs

This posterior-characterized modelling domain already presents specialized opportunities for analysis and inference. In particular, any properties or characterizations of $\alpha^{\{p\}}$ are semantically meaningful properties or characterizations of the ability of the experimental procedure to answer the query. For example, the variance (or entropy) of $\alpha(d)$ averaged across, maximized across, or otherwise weighted across potential outcomes d of $\mathcal{R}_{exp}$ represents the degree of certainty with which $\mathcal{R}_{qry}$ can be inferred from $\mathcal{R}_{exp}$.

For small examples, reduction of the modelling domain to the construction or characterization of this induced α posterior may be sufficient. However, it is still very a high-dimensional mathematical object in the general case, and often further concretization is useful as described next.

3.3 "Hard" Estimators and Classification

Practically, characterizing posteriors on $\mathcal{D}_{qry}$ is often much more information than we need.

β **example:**

For example, if the query program measures an observable value before and after an intervention, we may only care to know whether the difference in those two measurements is at least δ with probability at least $(1 - \epsilon)$ for some thresholds ϵ and δ[3].

This may be modelled as a function β that maps the complex space of distributions over $\mathcal{D}_{qry}$ to a much smaller (even binary) domain $\mathcal{A}$ representing answers to a particular question about that distribution. We can now attempt, instead of fully characterizing α, to just characterize some γ that directly provides an answer $\mathcal{A}$ for any observed value of $\mathcal{D}_{exp}$, but still corresponds to a theoretical composition of α and β. This relationship is depicted in figure 3.

Since γ seeks to provide a answer that is a concrete value, not a distribution of values, it is referred to as a "hard" estimate.

[3] for example $\epsilon = 0.05$ is a standard "p value" threshold

$$
\begin{tikzcd}
& \Theta \arrow[dl, squiggly, "\mathcal{R}_{exp}^{\{p\}}"'] \arrow[dr, "\mathcal{R}_{qry}^{\{p\}}"] & & \\
\mathcal{D}_{exp} \arrow[rr, dashed, "\alpha^{\{p\}}"] \arrow[rrr, dashed, bend right, "\gamma"'] & & \mathbb{P}_p[\mathcal{D}_{qry}] \arrow[r, "\beta"] & \mathcal{A}
\end{tikzcd}
$$

Figure 3: Complete pipeline

In its simplest form, when $\mathcal{A}$ is, for example, a boolean domain, a complete characterization of γ is simply a partition of all possible results of the experimental program into "yes" and "no" sets corresponding to the "correct" (as defined by the remainder of the model) answer $\mathcal{A}$ given that result.

When $\mathcal{A}$ is a largely, but still easily enumerable set, then inference of γ is an only slightly more general classification task on potential results of the experimental procedure.

3.4 γ-Approximacy

Notably, the complete modelling domain specified in figure 3 lends itself immediately to γ *-approximate* inference. Instead of directly computing $\mathcal{A}$ via γ that is equivalent to a composition of α and β, we can choose an *approximate* answer domain $\bar{\mathcal{A}}$, related to $\mathcal{A}$ through an *error function* $\mathcal{E}$ (for which $\bar{\mathcal{A}} = \mathcal{A}$ is an obvious and often-appropriate choice with natural choices of $\mathcal{E}$), and obtained via an approximate estimator $\bar{\gamma}$. This new model is depicted in figure 4.

$$
\begin{tikzcd}
& \Theta \arrow[dl, squiggly, "\mathcal{R}_{exp}^{\{p\}}"'] \arrow[dr, "\mathcal{R}_{qry}^{\{p\}}"] & & \\
\mathcal{D}_{exp} \arrow[rr, dashed, "\alpha^{\{p\}}"] \arrow[drrr, dashed, bend right, "\bar{\gamma}"'] & & \mathbb{P}_p[\mathcal{D}_{qry}] \arrow[r, "\beta"] & \mathcal{A} \arrow[d, dotted, leftrightarrow, "\mathcal{E}"] \\
& & & \bar{\mathcal{A}}
\end{tikzcd}
$$

Figure 4: γ-approximate pipeline

Here, the obligation to show γ as an exact composition of α and β is relaxed. Instead, we can reason about bounding the error $\mathcal{E}$ indcued between the exact γ and the approximate $\bar{\gamma}$. This is often sufficient.

Even more importantly, $\bar{\gamma}$ can be constrained to be a clearly *interpretable* family of models, such as deciison trees. This modelling scenario would thus suffice to show the conditions under which nuanced causal queries may be answered by a composition of a human-comprehensible data-generating process and a human-comphrenensible decision procedure on that data - like a "flowchart" style classification.

4 Certification Goals

The above information processing pipeline is meant to show how "procedure-to-claim" proofs decompose into a series of independently-verifiable algebraic transforms. This section does *not* attempt to show how real world choices of $\mathcal{R}_{qry}$, $\mathcal{R}_{exp}$, and β translate into particular claims inhabited by this proofs; for that, see section 5.

Given that the user either accepts *claim specifications* $(\mathcal{R}_{qry}, \mathcal{R}_{exp}, \beta)$ as given, or has internalized rationale for choosing them herself, the algebraic construction of the estimator γ whose semantics define the ultimate claim proceeds as follows:

1. Fix experimental and query programs $\mathcal{R}_{exp}$ and $\mathcal{R}_{qry}$, and a prior p on $\mathcal{U}$ (see figure 1)

2. Compute $\alpha^{\{p\}}$ - the ideal "soft" decision posterior estimate of $\mathcal{D}_{qry}$ (the theoretical result of $\mathcal{R}_{qry}^{\{p\}}$) given $\mathcal{D}_{exp}$ (the real result of $\mathcal{R}_{exp}^{\{p\}}$) (see figure 2)

3. Fix a "hard" decision $\mathcal{A}$ defined by a mapping $\beta : \mathbb{P}_p[\mathcal{D}_{qry}] \to \mathcal{A}$ - i.e. pick a concrete (non-probabilistic) property of distributions over $\mathcal{D}_{qry}$.

4. Compute γ - the ideal "hard" decision corresponding to the composition of computed α and chosen β (see fig 3).

5. (optional): Compute $\bar{\gamma}$ - an approximation of γ up to error $\mathcal{E}$ that may be more interpretable or more computationally feasible to evaluate, infer, or represent.

Each step of this operationalization lends itself to formal verification:

$\boxed{1 \to 2}$ Computing $\alpha^{\{p\}}$ directly (i.e. intensionally via raw distributions) is computationally feasible only for tiny domains. Formal semantics of $\mathbb{AIM}$ programs suffice to form and validate (extensional) algebraic representations of α, or verify the sound approximacy of other intensional representations.

$\boxed{2 \to 3}$ Various principles in theoretical statistic justify choices of β - for example, maximum liklihood, minimax loss, minimum variance unbiased estimators, information theoretic encoding schemes. Formal verification can show the correspondence between these principled estimators and domain-specific choices.

$\boxed{3 \to 4, (5)}$ Similarly to the $[1 \to 2]$ case, exact specification of γ is algebraically complex, and computationally infeasible. Verifiction of algebraic rewrites or computational approximation is highly valuable. In subdomains, *synthesis* of γ or $\bar{\gamma}$ up to certain bounded cost constraints may also be enabled.

Together, some or all of these certification goals would provide a meaingful improvement to the state of the art in verified implementation of causal inference.

Since any γ compatible with the framework in section 3 can be thus constructed, any such γ can also be thus verified. In this sense, the framework justifies itself as not just a construction of statistical estimators, but a *correctness by construction* doctrine for synthesis of statistical estimators.

5 Applying the Framework

The reader's first reaction to section 3 may center on appreciating from a distance that there's pretty math at play without really understanding what the mean means. This section attempts to remedy that.

6 Conclusion

Section 1 motivated the need to specialize the modelling tasks of general statistical/causal inference whenever possible, showing that when we focus on the particular situation of using one causal program to predict another in the context of a structural causal model, the vast majority of the complexity from the general case of overparametric inference can be eliminated.

Section 2 provided sufficient details of the modelling of causal programs to concretize further details of their analysis.

Section 3 provieded a framework for extending high level mathematical characterizations of causal programs to specifications of concrete "soft" and "hard" estimators for causal queries at varying levels of abstraction.

Section 4 began to discus the application of this framework to certified decision making procedures.

Finally, section 5 walked through a few example applications of the framework.

Together, this report begins to illustrate that the combination of general statistical inference, structural causal models, and concrete semantics for causal programs suffices to specify and certify efficient and explainable causal inferential procedures.